Screening and Auditing Data Leakage in Wearable Physiological Regression, with Application to PPG-Based Glucose and Blood Pressure

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Abstract*—*Objective: **Accuracies of physiological estimation from photoplethysmography (PPG) are often inflated by leakage that subject-wise splitting misses. We develop a method to screen for, detect and measure such leakage.** ***Methods:*** **The audit combines a label-construction detector, a copy probe, asserted split protocols, a leakage ladder, a null-anchored decision rule, and an a-priori screen for identity-leak inflation: label clustering (potential) times how well the inputs identify a subject (identifiability), computable before any model is trained. Components were validated on synthetic cohorts and applied to three glucose cohorts, a published glucose result, and two blood-pressure cohorts (219 and 556 subjects).** ***Results:*** **In a preregistered test on 20 new model–cohort pairs the screen flagged every leaking input set correctly and gave the inflation within a factor of two in 60% of pairs ($\rho = 0.68$). It placed the blood-pressure leak in demographic inputs: gradient boosting on age, sex, height and weight alone, without PPG, reached MAE 5.9 mmHg ($r = 0.93$) under a random split and no better than the mean with subjects separated. The label detector flagged a public glucose derivative without false alarms on 63 real subjects. A published glucose accuracy (MAE 2.96 mg/dL) was matched only when augmentation preceded the split. On 556 surgical patients a genuine calibration-free PPG signal remained (Δ -2.4 mmHg); glucose effects stayed below the reference noise.** ***Conclusion:*** **Identity leakage can be screened before training and audited after it.** ***Significance:*** **The open method lets wearable studies test for leakage.**

***Index Terms*—Photoplethysmography, data leakage, evaluation methodology, blood glucose, blood pressure, cross-validation, wearable sensing, reproducibility.**

I. Introduction

CUFFLESS estimation of blood pressure, glucose and other physiological variables from photoplethysmography (PPG) is an active engineering target, and reported errors often approach those of regulated devices [1]–[3]. Evaluation is the weak link. Record-wise splits of autocorrelated wearable signals inflate accuracy [4]–[6], and recent re-evaluations of PPG-based glucose estimation found that published models do not beat a mean predictor once subjects are separated [7], [8]. Subject-wise splitting is now the recommended remedy [9], [10], but it is not sufficient: leaks introduced while a dataset is *constructed* survive any split, and the magnitude of a leak depends on the model as well as on the protocol.

This paper treats evaluation as an engineering problem with a method that can itself be validated. Its contributions are:

G. Verma is with the Department of Electronics and Communication Engineering, [Institution Name] (e-mail: [author.email@institution.edu]). Code, frozen folds, integrity tests and all out-of-fold predictions: https://github.com/Namangarg484/IEEE-Gaurav-Verma.

1) **An audit method** (Sec. II) with six components: a detector for label-construction leaks with a block-scan variant for partial corruption, a copy probe for augmented or duplicated samples, split protocols whose no-leak guarantee is asserted in code before training, a leakage ladder that measures inflation mechanism by mechanism, a null-anchored decision rule with controls, and an *a-priori* screen for identity-leak inflation from label clustering and input identifiability, computable before any model is trained.
2) **Validation on known ground truth** (Sec. III): detector sensitivity, specificity and failure modes, the false-claim rate and power of the decision rule, and the leak-risk screen on synthetic cohorts and in a preregistered prospective test.
3) **Applications** (Sec. IV). For PPG→glucose the audit uncovers a previously unreported label-replication artefact in a public derivative, bounds what three cohorts support, and re-audits a published result, matching its accuracy only under a copy leak. For PPG→blood pressure, on a 219-subject cohort and on 556 surgical patients, it recovers a genuine signal, measures how much a random split inflates it and through which inputs, and quantifies how often a signal of that size would be detected in cohorts as small as the glucose cohorts.

II. Audit Method

A. Setting and leak taxonomy

A dataset is a set of windows w, each with a signal segment, a reference label y, a *record* (one independent reference measurement), a *subject* and, for longitudinal data, a time t. A split protocol produces folds $(\mathcal{T}_f, \mathcal{E}_f)$. We distinguish six leak levels (Table I): window, record, subject and temporal leaks arise from the split; copy leaks arise when augmented or duplicated samples are created before it; label-construction leaks arise when a table pairs signal rows with labels so that a label no longer belongs to the signal it accompanies. Kapoor and Narayanan's taxonomy [11] covers the first four and copies under its duplicates type; label construction is closest to duplicates but acts on the label column.

B. Label-construction detector

A cross-join of a subject's k reference values with its signal rows produces a label column with $y_i = y_{i-k}$ for almost all i, whereas a genuine table is record-structured (runs of one

TABLE I
LEAK LEVELS, THE PROTOCOL THAT REMOVES EACH, AND THE AUDIT CHECK.

Leak level	Mechanism	Protocol	Audit check
Window	overlapping windows of one record on both sides	record-wise or stricter	set intersection of windows
Record	records of one subject on both sides	subject-wise	set intersection of records
Subject	identity learnt, label = subject mean	subject-wise	set intersection of subjects
Temporal	neighbouring readings across the cut	chronological + guard gap g	min. time distance $> g$
Copy	augmented / duplicated copies of one sample on both sides	augment after splitting	copy probe (1-NN) must not beat the null
Label construction	labels cycled over signal rows (cross-join)	none (pre-split)	cycle detector, global and block scan

label per record). For a label column $y_{1:n}$ we compute the lag-ℓ match fraction $\phi(\ell) = \frac{1}{n-\ell}\sum_{i>\ell} \mathbb{1}[y_i = y_{i-\ell}]$ for $\ell = 1, \ldots, \ell_{\max}$ and report the smallest ℓ with $\phi(\ell) > \tau$ ($\tau = 0.98$). $\ell = 1$ indicates runs (genuine); $\ell \geq 2$ indicates cycling. Because a corrupted stretch covering a fraction p of a subject's rows is diluted to $\phi(k) \approx p + (1-p)(1-k/m)$ for run length m, the whole-column test misses partial corruption; the *block scan* applies the same test to contiguous blocks of b rows (stride $b/2$). On a record-structured column any block is a single run or crosses a run boundary, so $\phi(1) \geq 1 - 1/b > \tau$ whenever $b > 1/(1-\tau)$ and $b > \ell_{\max}$: the block scan cannot raise a false alarm on genuine labels stored in record (or acquisition) order, which the test therefore requires. It cannot detect a cross-join whose rows were subsequently shuffled; when record identifiers exist, a complementary check that every record carries exactly one label covers that case.

C. Split protocols with asserted guarantees

Each protocol declares the leak level it removes (Table I). Before any model is trained, the folds are checked by exact set arithmetic on window, record and subject identifiers and, for the temporal level, by the minimum distance between a test reading and any training reading of the same subject. A violation aborts the run. Folds are frozen to files with a fingerprint so a third-party model is scored on identical folds.

D. Leakage ladder

The same model is evaluated under a sequence of protocols that leak progressively less (random windows $\rightarrow$ random records $\rightarrow$ subjects; or random readings $\rightarrow$ chronological $\rightarrow$ chronological with guard gap). The difference in a metric between adjacent rungs measures the inflation due to one mechanism, separately from the others.

E. Null-anchored decision rule and controls

Window predictions are averaged per record. Every model is compared with the null appropriate to the use case: the training mean for cross-subject estimation; each subject's training mean, a time-of-day model and persistence for within-subject tracking. The paired MAE difference Δ is bootstrapped by resampling *subjects* with replacement (5 000 draws for the cohort analyses, 2 000 inside the simulations, the re-audit and the VitalDB grid), and a model is declared better than the null only if the 95% interval of Δ lies entirely below zero. A model trained with several seeds is judged on its seed-ensemble prediction (mean of the seeds' predictions per record); per-seed results are reported alongside. One comparison per cohort is pre-specified as primary; the others are exploratory. Positive controls (quantities that must be recoverable, e.g. heart rate) and negative controls (permuted labels, noise input) are run through the identical pipeline, and the reference device's own error (ISO 15197 meter, CGM MARD) is reported as a noise floor. A *copy probe*, a 1-nearest-neighbour regressor on the raw (z-scored) signal, is run wherever copies are plausible. It beats the null when a test sample has a close raw-signal neighbour with a similar label in training: an augmented or duplicated copy, or, when labels are subject-specific, a near-identical recording of the same subject. It is a targeted check for copies, not a general leak detector: it misses time-shifted overlapping windows of quasi-periodic PPG (Sec. III) and identity leaks that act through features, both of which the ladder measures.

F. A-priori leak-risk screening

The ladder measures inflation after the fact, for one trained model, as does the permutation test of Chaibub Neto *et al.* [12], which detects whether a trained classifier has learnt subject identity. We also estimate, before any target model is trained, how much a split that admits a subject's other samples into training can lower the error. The *potential* $P = \mathrm{MAE}_{\text{null}} - \mathrm{MAE}_{\text{oracle}}$ is the error removed by a subject oracle that predicts each test sample by the mean label of the same subject's training samples; it depends only on the labels and the split, and grows with the between-subject share of label variance (intraclass correlation, ICC). The *identifiability* $a(X)$ is the accuracy with which a 1-nearest-neighbour classifier on the standardised inputs X recovers the subject of each test sample under the same split. Assume that (i) the model carries no genuine signal and neither over- nor under-fits new subjects, (ii) it scores like the oracle on samples whose subject it recognises and like the null on the rest, and (iii) it can resolve a single subject, i.e. it does not average over more samples than a subject contributes. Then the leaky error is $a\,\mathrm{MAE}_{\text{oracle}} + (1-a)\,\mathrm{MAE}_{\text{null}}$ and the predicted inflation is

$$\hat{I}(X) = a(X)\,P. \tag{1}$$

If the honest result is available, replacing $\mathrm{MAE}_{\text{null}}$ by the honest model error drops assumption (i): $\hat{I}_2 = a\,(\mathrm{MAE}_{\text{honest}} - \mathrm{MAE}_{\text{oracle}})$. $a(X)$ is computed on the inputs the model receives, because auxiliary inputs such as demographics can identify subjects without carrying information about the target. Eq. (1) covers identity leakage only; neighbours in time add to it (Sec. IV-D). It is the analogue, for subject-clustered physiological data, of nearest-neighbour diagnostics of optimistic cross-validation in spatial modelling [13], [14].

III. VALIDATION ON KNOWN GROUND TRUTH

Detector. We generated label columns for 20 subjects, corrupted a fraction (10–100%) of them with cross-joins of

cycle length $k \in \{2, 3, 5, 7, 10\}$ over 25, 50 or 100% of their rows, and left the rest record-structured. Fig. 2a shows subject-level sensitivity. Both detectors had 100% specificity in every condition. The whole-column test detected fully corrupted subjects (sensitivity $\geq$97%) but missed partial corruption for long cycles (minimum 0%); the block scan detected every corrupted subject for $k \geq 3$ (100%) and at least 97% overall, the residual misses at $k = 2$ being draws in which the two reference values coincide and the cycle is invisible. On the label columns of 63 real subjects from three clean cohorts it raised 0 alarms; on the public glucose derivative of Sec. IV-B it flagged 21 of the 21 patients with more than one reference value. The clean synthetic columns are record-ordered by construction, so their specificity follows from Sec. II-B; we therefore also stressed the two assumptions. A cross-join whose rows were shuffled was detected in 0% of draws, and a genuine table stored window-major (window 1 of every record, then window 2, …) raised a false alarm whenever it had at most 30 records; the same tables in record order raised none. The detector is therefore valid only on rows in record or acquisition order.

Decision rule. We generated 200 synthetic cohorts per condition (16 subjects, 12 records of three 5-s windows each). Each label is a subject offset (SD 20 mg/dL, unlearnable from the signal) plus g times the heart-rate deviation encoded in the waveform plus reference noise. Each subject also has a pulse-shape fingerprint unrelated to the label, so a flexible model can learn identity, and hence the offset, whenever a protocol lets a subject's windows into training. Fig. 2b shows how often the rule declared a ridge or a gradient-boosting (GBM) model better than the train-mean null (4-fold subject-wise folds; the leaky arm also contains overlapping windows). With no signal ($g = 0$, 1000 additional cohorts) the honest false-claim rate was 0.4% for ridge (exact 95% CI 0.1–1.0%) and 0.5% for GBM (0.2–1.2%), below the nominal one-sided 2.5%: the rule is conservative. Under the leaky protocol it rose to 15.8% (ridge) and 99.9% (GBM), the GBM *appearing* to improve on the null by -9.3 mg/dL with no signal present: the rule is valid only together with an asserted protocol. With signal present the honest rule's power rose with g, to 96% (ridge) and 93% (GBM) at $g = 4$, but was only 6% and 2% at $g = 1$: with 16 subjects small effects go undetected. The generator has no confounder linking a nuisance variable to the label, a case the rule cannot distinguish from signal and the controls must address.

Leak-risk screening. We varied the between-subject label SD (5–40 mg/dL) and the strength of the subject fingerprint (none to full) independently, 128 cohorts in all (identifiability 4–73%), computed $\hat{I}$ from the labels and inputs alone, and then measured the inflation of gradient boosting. $\hat{I}$ correlated with it ($r = 0.97$, Spearman $\rho = 0.94$; Fig. 1a), but this agreement is partly fortuitous: gradient boosting was worse than the null on new subjects (median honest gain -2.1 mg/dL), a penalty that makes up about 57% of the observed inflation and that assumption (i) excludes. Against the leaky gain over the null, the quantity Eq. (1) models, the correlation is 0.87. The two-stage form $\hat{I}_2$, which uses the honest error, accounts for the penalty ($r = 0.99$; median observed/predicted 1.13; 80% of cohorts within a factor of 1.5).

Copy probe. On the Kermani–Esmaeili random-window rung, where overlapping windows of one recording cross the split, the Euclidean probe did not beat the null (Δ +0.1 mg/dL, [-2.4, +2.3]) and a shift-invariant variant (maximum normalised cross-correlation) only approached it (-2.7, [-5.3, +0.03]): quasi-periodic pulses correlate with any other pulse at some lag, so time-shifted copies do not stand out. The probe is decisive for augmented copies (Sec. IV-C) and is not a substitute for the ladder.

IV. Applications

A. Data and models

Glucose. Kermani–Esmaeili finger PPG [15] (67 10-s recordings, 23 subjects, fingerstick reference); BIG IDEAs wrist BVP with Dexcom G6 CGM [16], [17] (16 subjects, 28867 readings, HbA1c 5.3–6.4%); and STUBA ePPG [18] (125 records, 24 subjects, 3 with type-1 diabetes). **Blood pressure.** PPG-BP [19]: 219 subjects, one cuff reading and three 2.1-s finger-PPG segments each; VitalDB [20]: 10-s segments with beat-wise arterial-line labels from surgical cases (Sec. IV-D). Signals are band-passed 0.5–8 Hz, resampled to 64 Hz and z-scored per window (5 s for glucose). Models: nulls; ridge and gradient boosting on nine hand-crafted PPG features, with and without demographics; a 1-D CNN, 2-D wavelet-scalogram CNN, Takens-embedding PointNet and their cross-attention fusion [21]–[23]; re-implemented ResNet-1D and CNN–LSTM [2], [24]; and ridge probes on frozen embeddings of three PPG foundation models (PaPaGei-S [25], Pulse-PPG [26], AnyPPG [27]). Networks use one shared configuration (AdamW, fixed budget, no early stopping on held-out data) and three seeds, except on VitalDB, where every model was run once. Protocols: leave-one-subject-out (fingerstick cohorts), chronological 70/30 with a 1-h guard gap (BIG IDEAs; plus a personalised arm trained on the residual to each subject's training mean), 5-fold subject-wise (PPG-BP, VitalDB), and for VitalDB also random-segment and a calibration-based personal split (first half of each case trains, second half tests). Full tables are in the supplement.

B. PPG → glucose

Label-construction leak. A widely shared Kaggle derivative of the Kermani–Esmaeili recordings [28] cycles each patient's reference values row by row (Fig. 3), which the detector flags for every patient with more than one value. Every window of that file inherits the patient mean, so a per-patient oracle scores MAE 0.03 mg/dL with $r = 1.00$ under a random split, and a correct subject-wise split cannot reveal the problem. An earlier, unpublished draft of this work trained on the file and reported MAE 6.89 mg/dL; those numbers are not carried forward.

Ladder. On the raw recordings the 1-D CNN scores 10.4±0.2 mg/dL ($r = 0.66 \pm 0.05$) under random windows, but 13.5±0.5 and 13.4±0.5 once recordings or subjects are separated, against a null of 14.7 (supplement, Fig. S1): on this cohort window-neighbour leakage, not identity leakage, produces the inflation, because most subjects contribute one or two recordings. On BIG IDEAs the ladder is nearly flat (16.1,

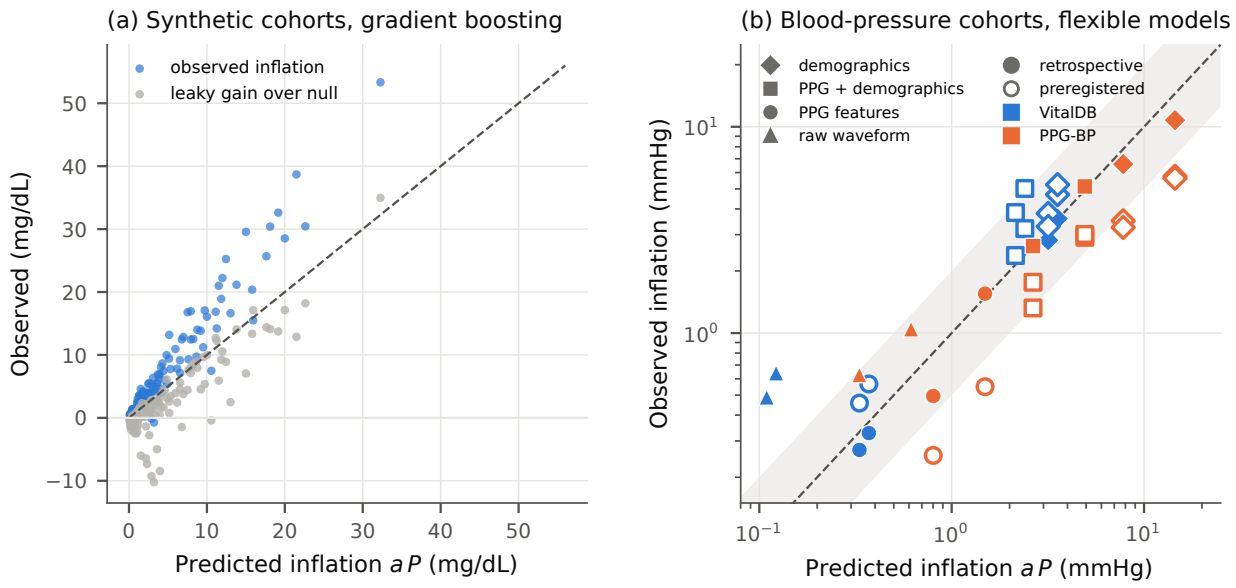


Fig. 1. Leak-risk screening. (a) Synthetic cohorts: predicted inflation $\hat{I} = a\,P$ (Eq. 1) against observed inflation (honest minus leaky MAE, blue) and against the leaky gain over the null (grey). (b) Blood-pressure cohorts, flexible models, log scales: filled markers were run before the predictor was formulated (retrospective); open markers are model types whose predictions were committed publicly before they were run (prospective). Dashed: identity; shaded: within a factor of two.

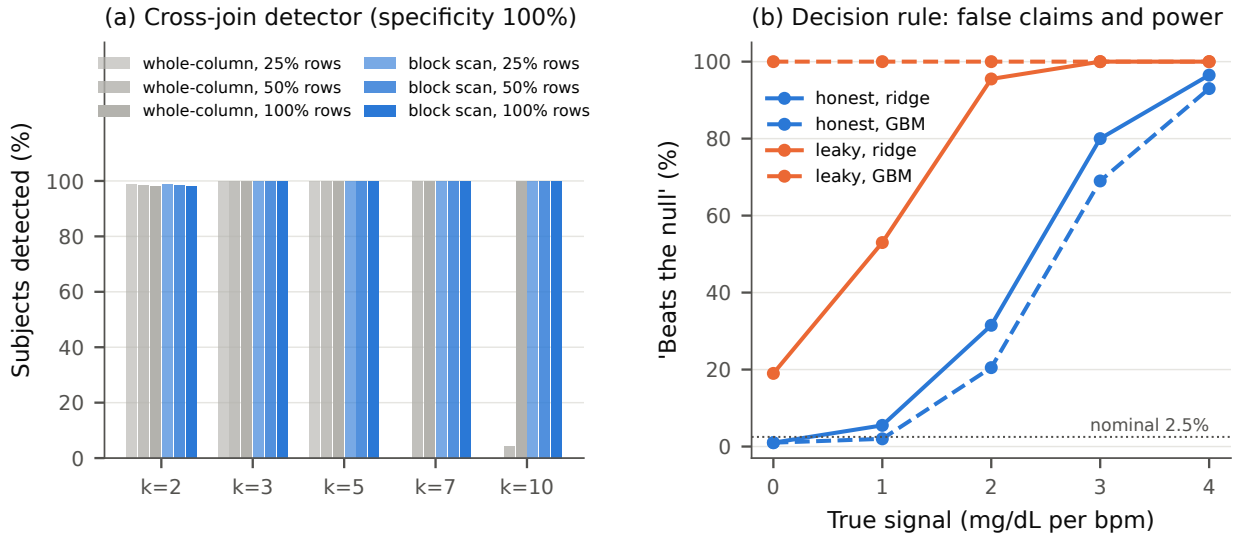


Fig. 2. Validation on known ground truth. (a) Subject-level sensitivity of the cross-join detector, whole-column versus block scan, by cycle length k and fraction of corrupted rows; specificity was 100% throughout. (b) Rate at which the decision rule declares a model better than the null, versus the true learnable signal g (mg/dL per bpm), under the honest subject-wise and the leaky random-window protocol; dotted line: nominal one-sided 2.5%.

16.9, 16.9 mg/dL) and every rung stays above the subject-mean null.

Bounds. Table II summarises the honest protocols. On the fingerstick cohort the pre-specified primary comparison (1-D CNN) gives Δ -1.4 mg/dL (95% CI [-3.0, -0.03]). By the decision rule this is a positive result, but a fragile and small one: the interval clears zero by a few hundredths, only 1 of 3 seeds clear it individually, the edge shrinks to 14.4±0.2 vs. 14.6 under 5-fold splitting, and the effect is a fifth of the reference meter's own error (7.1 mg/dL). On BIG IDEAs every population model is worse than each subject's own mean; with the personalised arm the best model, a Pulse-PPG probe, reaches Δ -0.3 mg/dL ([-0.5, -0.02]), one of 2 of 41 intervals below zero and an order of magnitude below the CGM noise floor (10.5 mg/dL); median within-subject correlation stays below 0.2 for every model. On STUBA no model beats the null. No PPG model detects any of the 68+34 hypoglycaemic references. Positive controls behave as required: heart rate is recovered from the same windows ($r = 0.74 \pm 0.01$), while permuted labels and noise inputs return to the null.

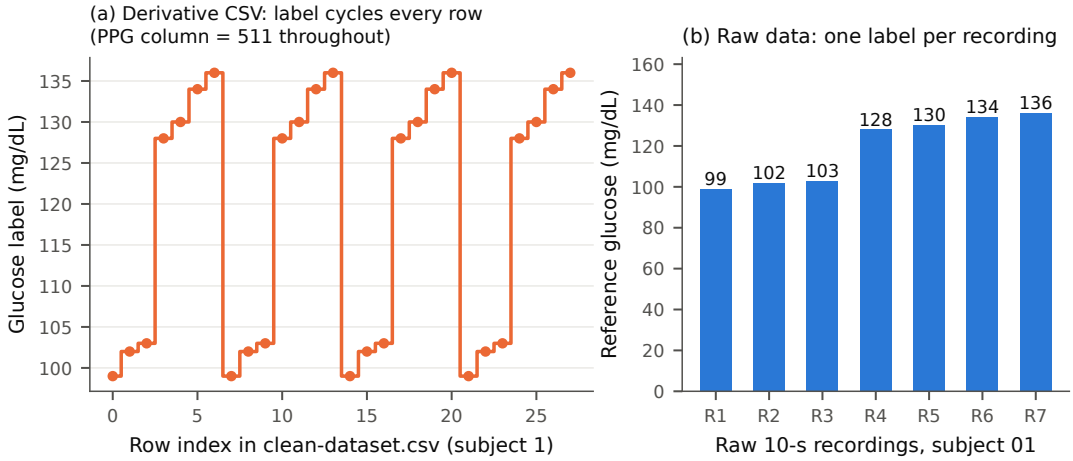


Fig. 3. Label-construction leak in a public glucose derivative. (a) The label of subject 1 cycles through the subject's seven reference values on consecutive rows. (b) The raw recordings of that subject carry one reference each.

TABLE II
PPG→GLUCOSE UNDER THE HONEST PROTOCOLS (SELECTED ROWS; ALL 41 COMPARISONS IN THE SUPPLEMENT). MAE IN MG/DL; Δ: PAIRED MAE DIFFERENCE VS. THE NULL WITH 95% SUBJECT-CLUSTER CI. ⋆: PRE-SPECIFIED PRIMARY COMPARISON.

Cohort	Model	MAE	r	Δ [95% CI]
K–E (LOSO)	Train mean (null)	14.7	–	–
	Ridge, PPG features	14.6	-0.03	-0.1 [-0.6, +0.6]
	1-D CNN ⋆	13.3	0.32±0.02	-1.4 [-3.0, -0.03]
	1D+2D+3D fusion	14.7	0.13±0.05	-0.1 [-1.5, +1.2]
	Pulse-PPG probe	14.2	0.14	-0.5 [-1.7, +0.8]
BIG IDEAs (chrono.)	Subject mean (null)	15.3	–	–
	1-D CNN	16.9	0.21	+1.7 [+0.5, +3.1]
	1D+2D+3D fusion	16.7	0.25	+1.4 [+0.2, +2.9]
	1-D CNN, residual	15.0±0.0	0.44±0.00	-0.2 [-0.7, +0.2]
	Pulse-PPG, residual	15.0	0.44	-0.3 [-0.5, -0.02]
STUBA (LOSO)	Train mean (null)	43.5	–	–
	1-D CNN	49.7±1.3	-0.15±0.01	+6.1 [+2.3, +7.6]
	PointNet (Takens)	43.2	-0.13	-0.3 [-1.3, +1.9]
	AnyPPG probe	43.5	0.04	+0.0 [-0.7, +1.1]

C. Re-audit of a published claim

Soliman *et al.* [1] report MAE 2.96 mg/dL, $R^2 = 0.97$ and 100% Clarke A for a CNN–GRU on the same 67 Kermani–Esmaeili recordings. The paper states a 0.5–8 Hz band-pass, downsampling to 30 Hz, Gaussian-noise augmentation from 67 to 269 samples, shuffling and 10-fold cross-validation. Its wording points both ways on the order of augmentation and split: noise is added "to the training PPG signals", yet "the sample numbers are increased from 67 to 269", a count of the whole dataset. The split unit, the noise levels and the layer sizes are not stated, and the headline values do not match the paper's own per-fold table (fold MAE 0.77–3.82 mg/dL). The re-audited result is an arXiv preprint. We reproduced the stated elements with a re-implementation of the described architecture (two Conv1D branches and a GRU branch, each followed by three dense layers; the GRU reads the sequence decimated to 10 Hz for speed), generated three noisy copies per signal (67→268 samples; noise SD 0.05, 0.10 and 0.20 of the signal SD), and resolved the unstated choices both ways (Table III). Early stopping uses a validation set drawn like the test set: samples in A, whole recordings or subjects, held out before augmentation, in B and C. Alongside the null we ran the copy probe of Sec. II-E.

With augmentation before a sample-level split (protocol A), the re-implementation matches or exceeds the claim: MAE 1.14±0.38 mg/dL, $R^2 = 0.95 \pm 0.05$, 99±1% Clarke A, and it does so at half and double our noise level (MAE 0.74 and 1.52). The memoriser does even better (MAE 0.13±0.19, $R^2 = 0.99 \pm 0.01$), because every test sample has noisy siblings of the same recording in training. Confining augmentation to the training folds and splitting by recording (B) or by subject (C) removes the copies, and the same network falls to 15.9±0.5 and 16.2±0.5 mg/dL against nulls of 14.7±0.0 and 14.5±0.0. Under the subject split the seed-ensemble does not beat the null (Δ +1.1 mg/dL, [-0.9, +3.1]); its point estimate is worse than the null in every seed, and in 1 of 3 seeds significantly so. The copy probe, now without copies, is worse still. The audit thus attributes the published accuracy to a copy leak created by augmenting before splitting, which the reported metrics cannot reveal and which any split that keeps a recording's copies together removes. We cannot verify the authors' exact pipeline, whose unstated choices we had to resolve; the memoriser shows that protocol A admits near-perfect scores without any physiological signal.

D. PPG → blood pressure

PPG-BP. Under the honest subject-wise split the 1-D CNN (seed ensemble) estimates systolic pressure with MAE 13.3 against 16.2 mmHg for the null (Δ -2.9, [-4.4, -1.5]; $r = 0.54 \pm 0.01$), and ridge on PPG features alone also clears the null (Δ -1.1, [-1.7, -0.6]) (Table V). On this cohort, however, age, sex, height and weight alone do nearly as well (Δ -2.4, [-3.5, -1.3]), and adding the PPG features to them does not help (13.8 vs. 13.9 mmHg): PPG-BP shows that the rule detects blood-pressure information, not that the information is specific to the waveform. Diastolic pressure is weaker; only the CNN clears the null (Δ -0.8, [-1.5, -0.2]).

TABLE III
RE-AUDIT OF A PUBLISHED PPG→GLUCOSE CLAIM [1] ON THE KERMANI–ESMAEILI RECORDINGS, 10-FOLD. MAE IN MG/DL, CLARKE A IN %; MEAN ± STD OVER 3 SEEDS. 1-NN: COPY PROBE. PER-SEED INTERVALS IN THE SUPPLEMENT.

Protocol	Model	MAE	R^2	A
Published [1]	CNN–GRU	2.96	0.97	100
A: augment, then split samples	Null	14.5±0.0	-0.01±0.00	85±0
	1-NN	0.13±0.19	0.99±0.01	100±0
	CNN–GRU	1.14±0.38	0.95±0.05	99±1
B: split recordings, augment train	Null	14.7±0.0	-0.04±0.01	85±0
	1-NN	18.5±0.2	-0.84±0.01	66±0
	CNN–GRU	15.9±0.5	-0.31±0.11	79±3
C: split subjects, augment train	Null	14.5±0.0	-0.02±0.00	84±0
	1-NN	18.1±0.2	-0.78±0.01	69±1
	CNN–GRU	16.2±0.5	-0.40±0.12	78±4

A random split over segments places a subject's other segments, and hence its label, in training. Ridge barely changes (13.6 vs. 13.8 mmHg); gradient boosting on PPG features alone improves from 14.5 to 13.0, and with demographics, which nearly identify a subject, from 14.6 to 9.4 mmHg (r 0.44 to 0.82). Inflation thus grows with model capacity and with quasi-identifiers among the inputs, and has to be measured for the model being reported. The copy probe does not beat the null under either split (Δ 0.6, [-0.9, +2.1] on random segments): segments of one recording start at different phases of the pulse and are not copies.

VitalDB. PPG-BP has one reading per subject, which limits identity leakage to three segments. We therefore drew 600 random surgical cases of VitalDB [20], an open source behind PulseDB [29] and many cuffless-BP results, and kept 21412 quality-controlled 10-s finger-PPG segments from the 558 cases (556 patients) that yielded any, each labelled with the median beat systolic peak and diastolic trough of the arterial line. Benchmarks that enforce subject-wise folds exist [30], but segment-level splits remain common [4]. Table IV gives the systolic results (diastolic in the supplement). Here demographics carry no information (Δ 0.0 mmHg), yet with patients separated the 1-D CNN beats the null (Δ -2.4, [-2.8, -2.1]; $r = 0.52$): a calibration-free signal carried by the waveform.

The leaky random-segment split, which places minutes-apart segments of the same surgery on both sides, leaves ridge unchanged (15.3 vs. 15.3 mmHg) and gradient boosting on PPG features nearly so (14.9 vs. 14.5), but gradient boosting with demographics falls from 15.3 to 12.1 mmHg and its correlation doubles (0.33 to 0.67). Demographics that predict nothing across patients identify them within the split, and the leak runs almost entirely through them. It also reverses the ranking of the two flexible models: on the leaky split gradient boosting beats the CNN (12.1 vs. 13.1), with patients separated the CNN wins (13.8 vs. 15.3). The copy probe stays above the null under every split, so the leak is an identity leak through features, visible on the ladder but not to the probe. For diastolic pressure the leak brings gradient boosting with demographics to an error SD of 8.8 mmHg, close to the 8-mmHg criterion of ISO 81060-2, which we use only as a reference scale (cuffless devices fall under IEEE 1708 [31]); with patients separated its SD is 11.6 mmHg. A split error alone can make a model look nearly compliant with a device standard. Under the calibration-based protocol, where models learn each case's deviation from its own training mean, all models improve on that mean, but by little more than a millimetre of mercury (best: 1-D CNN, Δ -1.4 mmHg, [-1.8, -1.1]).

Leak-risk screening. The order of work matters here. The ladder first showed the demographic leak; Eq. (1) was formulated afterwards, and its agreement with the seven model types run before any prediction was committed is a retrospective check ($r = 0.99$, $\rho = 0.91$ over 16 flexible model–cohort pairs). We then committed predictions for five model types never run on these cohorts (random forest and k-nearest neighbours on demographics, random forest on PPG and demographics, extra-trees on PPG features, and a more regularised gradient boosting) to the public repository (commit `47d6ae2`) before running them. On these 20 prospective pairs agreement was moderate ($r = 0.66$, $\rho = 0.68$; median observed/predicted 0.85; 60% within a factor of two; Fig. 1b). The screening decision held without exception: every prospective model whose inputs included demographics (identifiable in 93–100% of cases; $\hat{I} \geq 2$ mmHg) inflated by at least 1.3 mmHg, and the PPG-only model ($a = 10\%$, $\hat{I} < 1.6$) by at most 0.6. The size errors have identifiable causes. On PPG-BP, models that average over at least five samples (forest leaves, $k = 5$) realised a third to two-thirds of the prediction, because a subject contributes only about two training segments (assumption iii). On VitalDB, models exceeded it, partly because they fit new patients worse than the null (assumption i), and, when PPG features are also input, because segments minutes apart in the same surgery add a temporal leak: an oracle that copies the nearest-in-time training segment of the same case has potential 7.0 mmHg against 3.6 for the subject oracle. Linear models, which cannot express per-subject offsets, realised at most 0.3 mmHg. The sharpest case is gradient boosting on age, sex, height and weight alone: on PPG-BP this PPG-free model scores MAE 5.9 mmHg with $r = 0.93$ under the random split, numbers that, reported alone, would suggest an accurate cuffless estimator, and 16.7 mmHg, no better than the mean, with subjects separated.

Power at glucose-cohort sizes. Drawing 23 VitalDB patients with three segments each (the Kermani–Esmaeili design) from the honest predictions, the rule detected the CNN's systolic signal in 46% of draws; with 16 patients and all their segments (the BIG IDEAs design) in 63%, and with 23 PPG-BP subjects in 28%. A signal of blood-pressure size would often be missed in the glucose cohorts, so their negative results bound the effect, by the minimum detectable effect, rather than prove its absence.

V. DISCUSSION

What the audit adds. Subject-wise splitting, the standard remedy, did not detect the largest problem we found: a label-construction leak in a public table, which the detector finds in seconds from the label column alone, provided the rows are in acquisition order. Nor does it address copies made before the split: the re-audited claim was matched only under

TABLE IV
VITALDB SYSTOLIC BP (556 PATIENTS, 21412 SEGMENTS, ONE SEED). MAE IN MMHG; r_w: MEDIAN WITHIN-CASE CORRELATION; Δ VS. THE PROTOCOL'S NULL (TRAIN MEAN; CASE MEAN FOR "PERSONAL", WHERE MODELS LEARN THE DEVIATION FROM IT) WITH 95% PATIENT-CLUSTER CI.

	Random segments		Subject-wise		Personal	
Model	MAE	r_w	MAE	Δ [95% CI]	MAE	Δ [95% CI]
Null	16.2	–	16.2	–	13.8	–
Copy probe	17.5	0.18	18.0	1.7 [+1.5, +2.0]	–	–
Ridge, demo.	16.2	–	16.2	0.0 [-0.04, +0.1]	–	–
Ridge, PPG	15.3	0.53	15.3	-0.9 [-1.1, -0.7]	13.1	-0.7 [-0.9, -0.5]
GBM, PPG	14.5	0.58	14.9	-1.3 [-1.6, -1.1]	12.8	-1.0 [-1.2, -0.7]
GBM, PPG+demo.	12.1	0.58	15.3	-0.9 [-1.3, -0.6]	12.6	-1.2 [-1.5, -1.0]
1-D CNN	13.1	0.72	13.8	-2.4 [-2.8, -2.1]	12.4	-1.4 [-1.8, -1.1]

TABLE V
PPG-BP (219 SUBJECTS): MAE (MMHG) UNDER THE LEAKY RANDOM-SEGMENT AND THE HONEST SUBJECT-WISE SPLIT; Δ VS. THE TRAIN-MEAN NULL WITH 95% SUBJECT-CLUSTER CI (HONEST SPLIT; CNN: SEED ENSEMBLE OF 3).

	Model	**Leaky**	**Honest**	**Δ honest [95% CI]**
SBP	Null	16.3	16.2	–
	Ridge, demo.	13.7	13.9	-2.4 [-3.5, -1.3]
	Ridge, PPG	15.0	15.1	-1.1 [-1.7, -0.6]
	GBM, PPG	13.0	14.5	-1.7 [-2.8, -0.6]
	GBM, PPG+demo.	9.4	14.6	-1.7 [-3.1, -0.3]
	1-D CNN	12.2±0.2	13.3	-2.9 [-4.4, -1.5]
DBP	Null	8.7	8.7	–
	Ridge, demo.	8.4	8.5	-0.2 [-0.5, +0.2]
	Ridge, PPG	8.6	8.7	0.0 [-0.2, +0.2]
	GBM, PPG	8.1	8.6	-0.1 [-0.6, +0.4]
	GBM, PPG+demo.	5.5	8.2	-0.6 [-1.3, +0.1]
	1-D CNN	7.2±0.1	7.9	-0.8 [-1.5, -0.2]

such a copy leak, which the copy probe exposes directly. The ladder separates mechanisms that are usually conflated: on the glucose recordings window overlap, not identity, inflated the result; on PPG-BP and VitalDB identity leakage inflated flexible models, mostly through demographic inputs that act as quasi-identifiers, and on VitalDB it reversed which model looked best. Because the inflation depends on the model and its inputs, a protocol cannot be certified once for all models; the ladder has to be run for the model being reported. The leak-risk screen turns part of this into a design check: label clustering and input identifiability, both computable before training, flagged correctly in a preregistered test which inputs would carry the identity leak, including demographics that carry no information about the target, and gave its size to within a factor of two in most cases. Identity confounding has been detected after training [12], and nearest-neighbour distances are used to diagnose optimistic cross-validation in spatial modelling [13], [14]; we are not aware of an earlier estimate of identity-leak magnitude made before training and tested prospectively. Under honest protocols the null-anchored rule kept its false-claim rate below nominal in simulation; on real data it declared the blood-pressure signals and, narrowly, the pre-specified Kermani–Esmaeili glucose comparison, an effect a fifth of the meter's own error. Ramesh *et al.* [7] released a re-implementation pipeline with participant-aware splits; the audit differs in asserting each protocol's guarantee in code, detecting construction and copy leaks, measuring inflation per mechanism, and validating its own decision rule.

What the applications show. A 2-s (PPG-BP) or 10-s (VitalDB) finger-PPG segment carries calibration-free information about systolic blood pressure, which on VitalDB is not explained by demographics. For glucose, every architecture tested, including frozen foundation models, showed at most a weak and protocol-dependent cross-subject edge and no measurable within-subject tracking, and no PPG model detected a hypo- or hyperglycaemic reading. Because a blood-pressure-sized signal would often be missed at the glucose cohorts' sizes, these are bounds on the effect in small, largely non-diabetic public cohorts, not proof that PPG carries no glucose information.

Recommendations. Before training, compute the leak risk $\hat{I} = a\,P$ for every input set; run a construction-leak check on every derived table, in acquisition order; augment only after splitting and run a copy probe; name the protocol and the leak it removes, and freeze folds; measure the ladder for the reported model, with and without quasi-identifying inputs; report a null and the reference noise floor next to every result, with subject-cluster intervals and a pre-specified primary comparison; and include positive controls.

Limitations. The re-audit reproduces the stated elements of a preprint's pipeline, not the authors' code; its unstated choices were resolved both ways, and only one claim was re-audited. VitalDB patients were anaesthetised and often received vasoactive drugs, so its blood-pressure dynamics differ from ambulatory use; its quality control has no arterial-damping check, missing demographics were mean-imputed, and its models were run with one seed. The detector targets order-preserving cycling; shuffled cross-joins need the record-label check, and other construction errors (e.g. shifted joins) need their own tests. The copy probe misses time-shifted copies of quasi-periodic signals. The synthetic validation is moderate in size and has no confounders. The leak-risk screen gives magnitude only to within about a factor of two, assumes the model can resolve single subjects, covers identity but not copy or temporal leaks, and was formulated after the retrospective cases; its identifiability term is a single-sample nearest-neighbour test, not monotone in the inputs, and underestimates what a network extracts from the raw waveform. Networks shared one configuration; encoder probes were linear and used one common preprocessing. The glucose cohorts contain few diabetic subjects and few hypoglycaemic references.

VI. CONCLUSION

We presented a method to predict and audit leakage in wearable physiological regression and validated its components on known ground truth, including their failure modes. Its a-priori screen, computed from label clustering and input identifiability before any model is trained, identified in a preregistered test which inputs carried an identity leak and estimated its size to within about a factor of two; a PPG-free model using the flagged demographic inputs looked like an accurate cuffless estimator under a random split. Applied to public PPG data it uncovered a label-construction leak in a glucose derivative, attributed a published glucose accuracy to a copy leak, measured how a random split inflates and reorders cuffless blood-pressure models on 556 surgical patients, mostly through

quasi-identifying inputs, while confirming a genuine waveform signal, and bounded the glucose information in small public cohorts. The implementation, frozen folds and all predictions are released so that any new claim can be audited the same way.

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